CHANDU PRASAD BHAIRAPU



AI Engineer

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ABOUT ME

Master's graduate in Artificial Intelligence with 3 years of professional experience in AI engineering and software development. Specialized in building AI products end-to-end, from prompts, tool calls, RAG and agent logic to evals, React and Python web apps. Most recently sole developer of an agentic knowledge-extraction system for the German energy sector, backed by daily AI-first development with Claude Code.

EXPERIENCE

Associate Generative AI Developer - Agentic Indexing Pipeline Engineer for German Energy Sector

Kniit GmbH

November 2025 – present Muldestausee, Germany

- Owned an AI product end-to-end as sole developer, from the idea to a working system: ingestion, chunking, prompt engineering and indexing, reporting directly to the CTO and working with SAP domain experts to understand what they needed.
- Designed the agent logic in LangGraph (entity-extraction, relationship-analysis, consolidation and reflection agents in parallel) for regulatory and technical documents; evals against an SAP-system-derived ground truth raised precision/recall from 33%/45% to 93%/90%.
- Engineered a deterministic, code-based pipeline for a second, high-volume source stream instead of LLM calls, cutting a $200,000 projected LLM budget to near-zero recurring cost while pushing accuracy toward 100%.
- Built a Model Context Protocol (MCP) service exposing the knowledge graph as tools with millisecond-latency retrieval, integrated into Claude Code with custom skills for agentic development workflows.
- Developed Expert Interface, a full-stack web app (React frontend, Python microservices backend) on Azure where domain experts validate and approve knowledge-graph changes.
- **Tech stack: Python, LangGraph, LangChain, LiteLLM, MCP, Claude Code, RAG, GraphRAG, React, FastAPI, Neo4j, MongoDB, Postgres, Docker, Kubernetes, Azure**

GenAI Developer (AI Consulting Intern) - Development and Deployment of an Agentic AI Workflow

Covestro Deutschland AG

December 2024 – April 2025 Leverkusen, Germany

- Worked with business stakeholders to understand their decision-making workflows, advised where agentic AI could help and translated their needs into the product approach.
- Designed and deployed an agentic AI assistant with LangGraph that routes requests across vector database queries, general inference and summarization; React frontend and Python/FastAPI backend, API tests with Postman.
- Containerized the stack with Docker and deployed it to AWS with CDK; versioned in GitLab with CI/CD pipelines.
- **Tech stack: Python, React, FastAPI, LangGraph, LangChain, Docker, AWS (CDK, EC2, Lambda), GitLab CI/CD**

GenAI Developer (Master's Thesis) - Evaluating the Performance of RAG Systems

Covestro Deutschland AG

May 2024 – November 2024 Leverkusen, Germany

- Automated evals of RAG generation and retrieval with RAGAS and PytrecEval against chemist-provided ground truth, on serverless AWS infrastructure for PrismaChat RAG evaluation (Lambda, SQS, DynamoDB) built with CDK.
- Benchmarked multilingual embedding models on over 1M chunks of internal chemical data and designed RAG pipelines (chunking, vector databases, indexing, fine-tuning models for retrieval).
- **Tech stack: Python, RAGAS, PytrecEval, LangChain, Hugging Face Transformers, PyTorch, AWS, Azure AI Search**

Data Scientist (Intern) – ML-Based Prediction of Prepolymer Synthesis Outcomes

Covestro Deutschland AG

November 2023 – April 2024 | Leverkusen, Germany

- Built and compared Random Forest, XGBoost and LSTM models to predict prepolymer synthesis outcomes from time-series process data, and delivered a Streamlit dashboard for chemists to simulate parameter changes before running physical batches.
- **Tech stack: Python, Scikit-Learn, XGBoost, TensorFlow/Keras, Pandas, NumPy, Streamlit**

Machine Learning Engineer - Electrical Vehicle Load Forecasting using Time Series Analysis

Cognizant Technology Solutions

September 2021 – March 2022 | Hyderabad, India

- Developed ARIMA, SARIMA, XGBoost regressor, FB Prophet and LSTM models for EV load time-series forecasting; the LSTM reached 96% test accuracy with strong performance on unseen data.

PROJECTS

Project: Voice Receptionist Agent | LiveKit | Deepgram | ElevenLabs | Python | React

- Real-time voicebot for a small business (Voice AI receptionist) with LiveKit, Deepgram speech recognition and ElevenLabs text-to-speech for conversational AI: conversation flows, prompts and tool calls (FAQ via RAG, take a message, book a callback, handoff to a human with a summary).
- Built spec-first with Claude Code against acceptance tests; golden-conversation evals in CI and a React dashboard that monitors latency per stage, tool calls and call outcomes.

Project: Spec-Driven Coding Agent Harness | Claude Code | MCP | Langfuse | GitHub Actions

- Turns a plain-language feature request into a testable spec (acceptance criteria as pytest tests) that Claude Code implements in agentic loops under a rule set (CLAUDE.md, custom skills, hooks blocking protected files and secrets); an MCP server gives the agent spec and task context.
- GitHub Actions gates every agent-written change with tests, static security analysis, dependency scanning and an agent eval suite (pass rate, cost, turns per task); traces go to Langfuse.

TECHNICAL SKILLSET

- **AI Products & Agents:** assistants and automations with LLMs, prompts, tool calls (MCP), RAG, GraphRAG, agent logic with LangGraph and LangChain, LiteLLM, evals (ground truth, RAGAS, PytrecEval), Hugging Face Transformers
- **AI-First Development:** Claude Code (custom skills, MCP integration, agentic coding workflows)
- **Full Stack:** Python, FastAPI, Python microservices, React (beginner), SQL, Postman
- **Data & Cloud:** Postgres, MongoDB, Neo4j, vector databases (Qdrant, Pinecone, Azure AI Search), Docker, Kubernetes, Azure, AWS (Lambda, SQS, DynamoDB, CDK), Git, GitLab CI/CD

EDUCATION

Master of Science in Artificial Intelligence — March 2022 – April 2025
Technische Hochschule Würzburg-Schweinfurt, Würzburg, Germany

Bachelor of Technology in Mechanical Engineering — August 2017 – July 2021
Minor in Computer Science & Engineering
G Pulla Reddy Engineering College, Kurnool, India

PUBLICATIONS & CERTIFICATIONS

- **[1]** Navigating the Future: An approach of Autonomous Indoor Vehicles
- Udemy - LLMs Mastery : Transformers & Generative AI

LANGUAGES

English (Business Fluent) | German (B1) | Hindi (Fluent) | Telugu (Native)